

Hossam Abu-elez Fathy

Machine Learning Engineer | Data Analyst | AI Student

Tossamabuelez@gmail.com • +201004937607 • Mansoura Dakahliya, Egypt

[LinkedIn](#) • [GitHub](#)

Profile

Computer Science and Artificial Intelligence student with hands-on experience in machine learning, data analysis, and software engineering. Built end-to-end AI projects involving neural networks, optimization algorithms, and full-stack applications. Seeking internships and entry-level opportunities in Machine Learning and Data Science.

Education

Artificial Intelligence Sciences at New Mansoura University, New Mansoura

October 2023 - June 2027

GPA: 3.5 / 4.0

Transact SQL queries using SQL server at Mahara Tech

January 2025 - June 2025

Experience

- Academic Experience

Internships

- Machine Learning Trainee - Creativa Innovation Hub, Mansoura Gained hands-on experience in machine learning fundamentals, including data preprocessing, model training, and evaluation using Python. Worked on practical ML tasks and applied key algorithms to real datasets under expert supervision.

Skills

Programming and Problem Solving: C++, Python (Pandas, NumPy, Scikit-Learn)

Machine Learning: Supervised & Unsupervised Learning

Data Analysis & Visualization: Plotly, Matplotlib, Seaborn

Database Management: SQL (PostgreSQL, MySQL)

Tools & Abilities: Jupyter Notebook, Visual Studio, Kaggle, Presentation, Communication, Teamwork and Collaboration

Volunteering

- Student Mentor (AI & Programming) supporting technology, helping students and beginners understand basic concepts and improve their technical skills.
- Tech Event Organizer Helped organize and coordinate AI and technology workshops for students, managing logistics and ensuring a smooth learning experience for all participants.

Projects

MedOptima AI Platform | AI Healthcare Decision Support System

Built a full-stack medical AI platform combining Neural Networks, Knowledge-Based Systems, and Genetic Algorithms. Achieved 99.46% test accuracy on a Kaggle dataset with 41 diseases and 117 symptoms. Developed REST APIs and an interactive dashboard for diagnosis, analytics, and treatment optimization.

Air Quality Prediction using Machine Learning 2024-2025

Developed a predictive ML model to forecast air pollution levels by integrating pollutant data with temperature and humidity. Performed data cleaning, feature engineering, and model training using Python, achieving reliable forecasting results and insightful visual analysis.

AI Optimization Research Project - FTT Algorithm 2025

Developed and analyzed a meta-heuristic optimization algorithm (Football Team Training - FTT) to solve supply chain optimization problems.

Airline Reservation System - Full Software Engineering Project 2023-2024

Designed system requirements, use-case diagrams, class diagrams, and sequence diagrams. (Built using OOP principles, after then enhanced with DB integration for efficient) Built the system architecture and applied UML and software engineering methodologies.

ALU 4-bit Digital Design Project 2024

Designed and implemented a 4-bit ALU supporting arithmetic and logical operations using digital logic and control signals.

Simulated and tested behavior using logic design tools.

Computer Architecture & Assembly - 8085 & 8086 2024

Built a strong understanding of low-level operations and instruction-level execution.

Applied assembly instructions in simulations and step-by-step trace analysis.

Languages

Arabic - Native

English - Proficient